

1 with  $51 \leq j \leq 100$

## EEML 2022: Causal Inference tutorial

Ex1 Dataset contains:

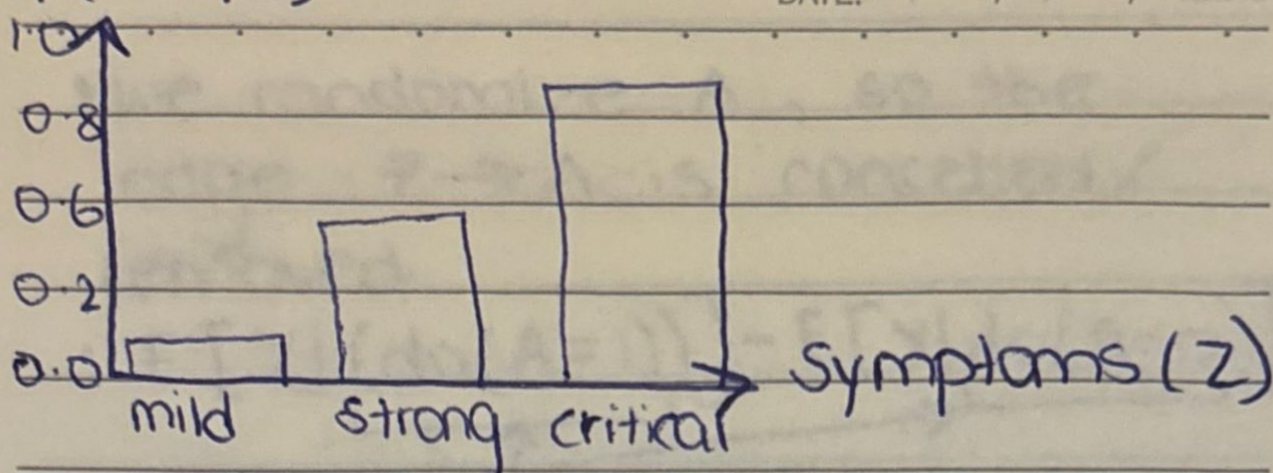
- $Z \rightarrow$  severity of symptoms ( $0 \rightarrow 2$ )
- $A \rightarrow$  administered treatment  
(treated is 1 & untreated is 0)
- $Y \rightarrow$  survival after one month  
(alive = 1 & dead = 0)
- Average treatment effect  $\rightarrow$   
chances of survival with taking  
vs. not taking the drug
- estimate it using:  
$$E[Y|A=1] - E[Y|A=0]$$

• when calculating  $\uparrow$  using the dataset we were provided, the ATE turned out to be a negative value  $\sim -0.11025$   
 $\rightarrow$  Drug is detrimental  
for survival, it is not effective.



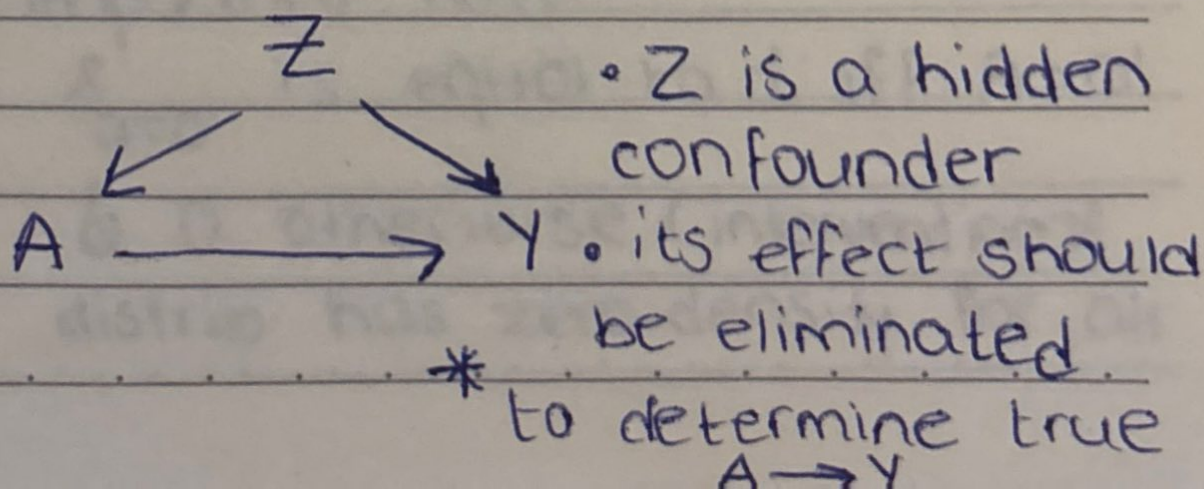
$$P(A=1|Z)$$

DATE. / /



- Graph represents the probability of receiving the drug w.r.t the symptoms
- As seen in the graph, the more "critical" your symptoms are the more likely you are to receive treatment
- Also observed that chances of survival decrease with the intensity of the symptoms Z.

current causal graph



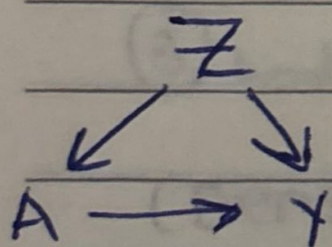


- we randomize  $A$ , so the edge  $Z \rightarrow A$  is cancelled / removed.

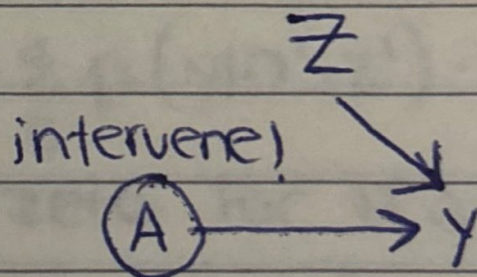
$$E[Y | \underbrace{do(A=1)}_{\text{intervening on } A \& \text{ setting its value forcibly}}] - E[Y | \underbrace{do(A=0)}_{\text{intervening on } A \& \text{ setting its value forcibly}}]$$

intervening on  $A$  & setting its value forcibly

- our initial assessment of the drug effectiveness using the ATE metric was biased due to hidden confounder  $Z$ .



$$P(a, y, z) = P(a|z) \cdot P(y|a, z) \cdot P(z)$$



$$P(a, y, z | do(A=a')) = P(y|a', z) \cdot P(z) \cdot \delta_{a=a'}$$

- Important note:

$\delta_{a=a'}$  is equal to 1 iff  $A=a'$

& 0 otherwise (interventional distrib. has zero density for all



values of  $a$  that don't correspond to the value of the interven.

$(a')$ .

- Extracting distribution of  $y$  in this interventional world so we marginalize out the other variables  $a$  &  $z$ :

$$\textcircled{1} \underline{P(y | do(A=a'))} = \sum_a \sum_z P(a, y, z | do(A=a'))$$

$$\textcircled{2} \underline{\quad} = \sum_a \sum_z p(y | a', z') \cdot P(z) \cdot \delta_{a=a'}$$

$$\textcircled{3} \underline{\quad} = 0 + \sum_z p(y | a', z') \cdot P(z) \cdot 1$$

(Density is zero for  $\forall a \neq a'$ )

$$\textcircled{4} \sum_z P(y | a', z) \cdot P(z)$$